SERVERLESS EXPENSE TRACKER STATICS WEB SERVER ON AWS

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3:30PM to 5:00PM

INDEX:

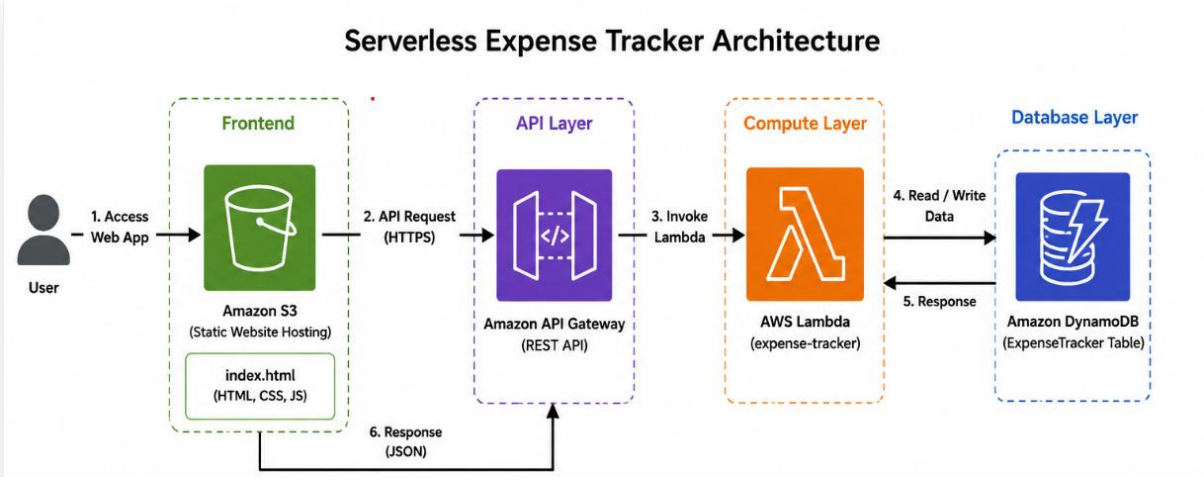
INTRODUCTION:

This project is a Serverless Expense Tracker Web Application developed using Amazon Web Services (AWS) via deployed using. The front-end is hosted via S3 static web. Backend database has connected with Lambda function (python 3.13 version) via API Gateway storing data handled by DynamoDB.

AWS SERVICES:

1. **DYNAMO DB:** Data is stored in tables using primary keys such as expense_id. And integrates easily with AWS services like Lambda and API Gateway.
2. **LAMBDA FUNCTION:** Create a function for backend server with (python 3.13) to fetch the static web site hosted on S3.
3. **API GATEWAY:** Create Rest API method to invoke through dynamodb.
4. **S3 bucket:** Hosted the public frontend web server via (HTML CSS AND PYTHON).

DIAGRAM:



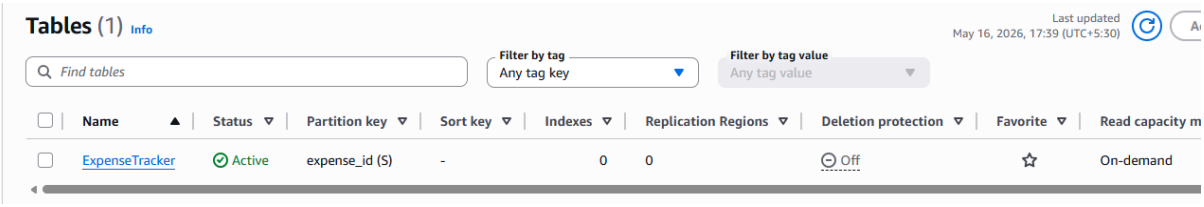
EXPLANATION:

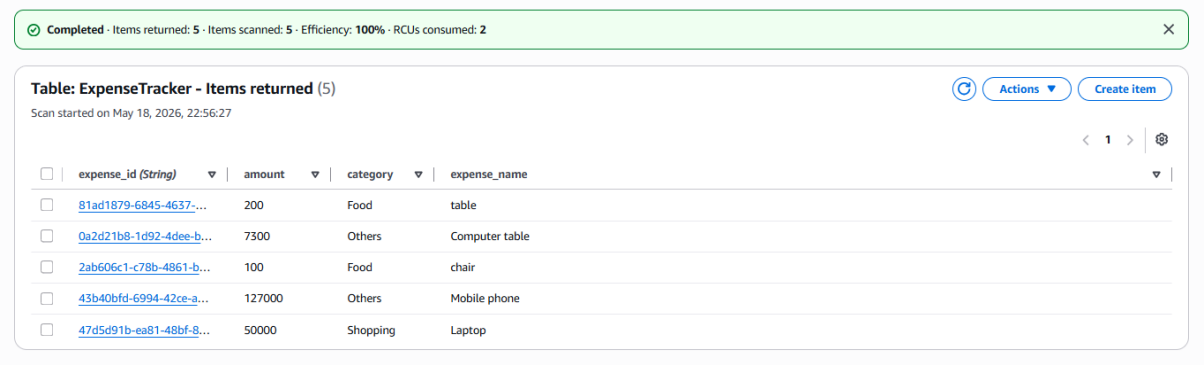
- Hosted the S3 static frontend website for expense tracker.
- Create a LAMBDA function and execute the python code to connect the DB table to store data.
- Created an API resources for expenses.
    - Create post method with Rest API and attached the lambda function and deploy.

INSTRUCTIONS:

1. DYNAMODB

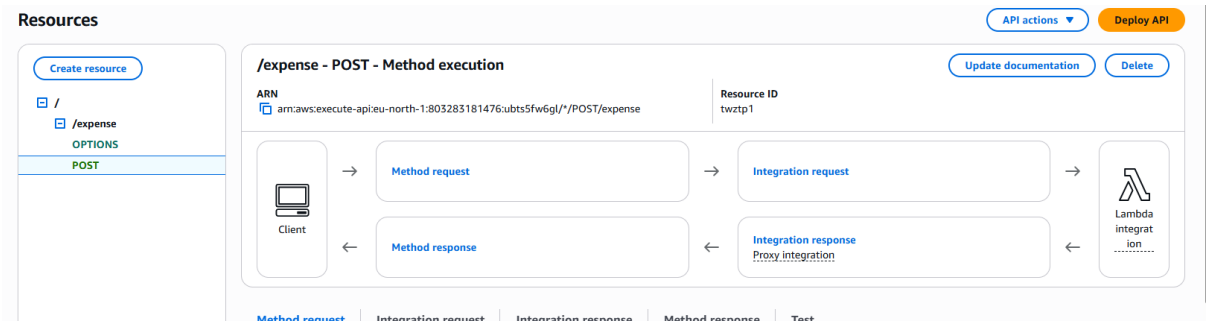
- Create a Dynamodb table for expenses
- Define partition key expensed_id





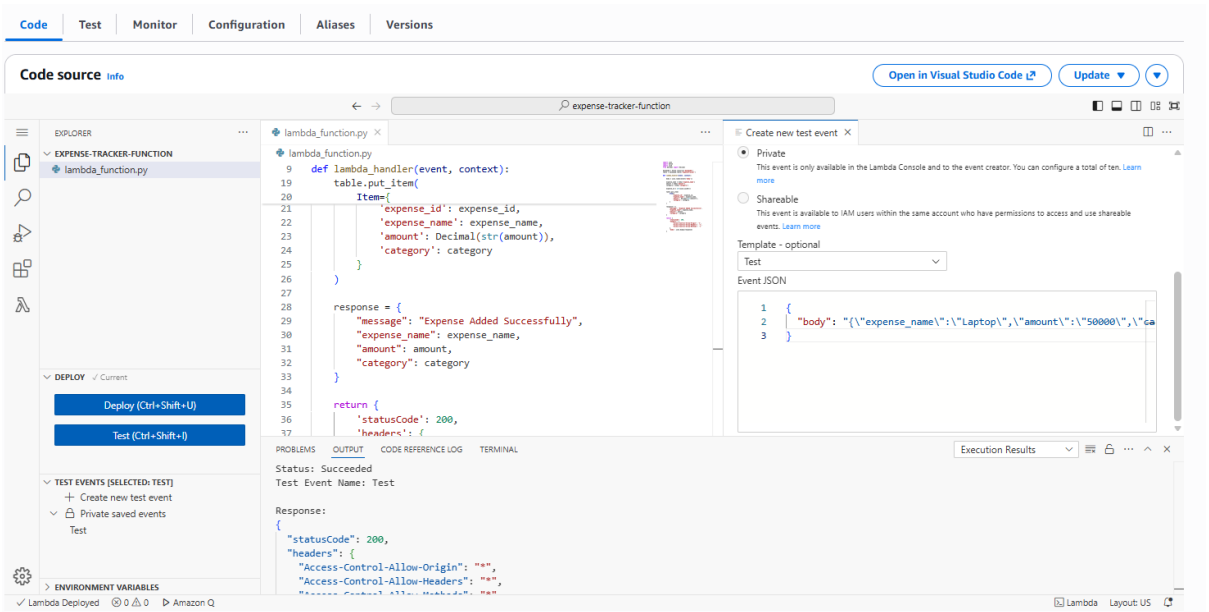
2. API GATEWAY:

- Create a Rest API in API Gateway.
- Add Resources and method
    - Post – post the data in Lambda.
    - Enable the lambda proxy integration to send all request.
- Enable CORS and deploy API, API URL has created.



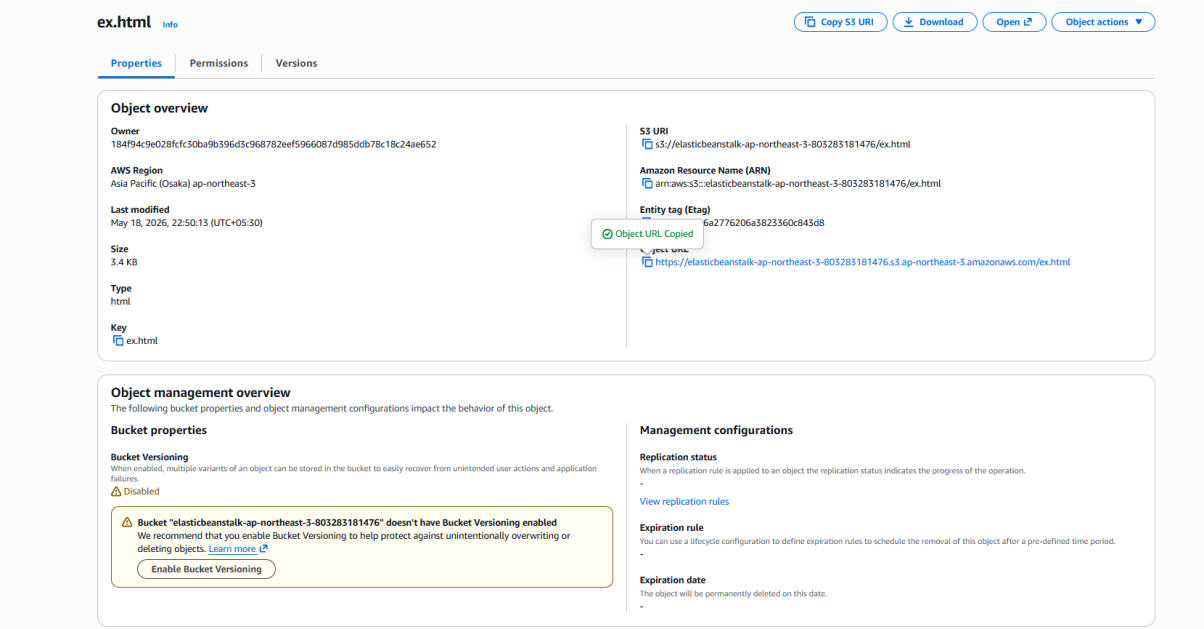
3. Lambda function

- Create a Lambda function deploys the python code.
- Enable the lambda proxy to get the request.

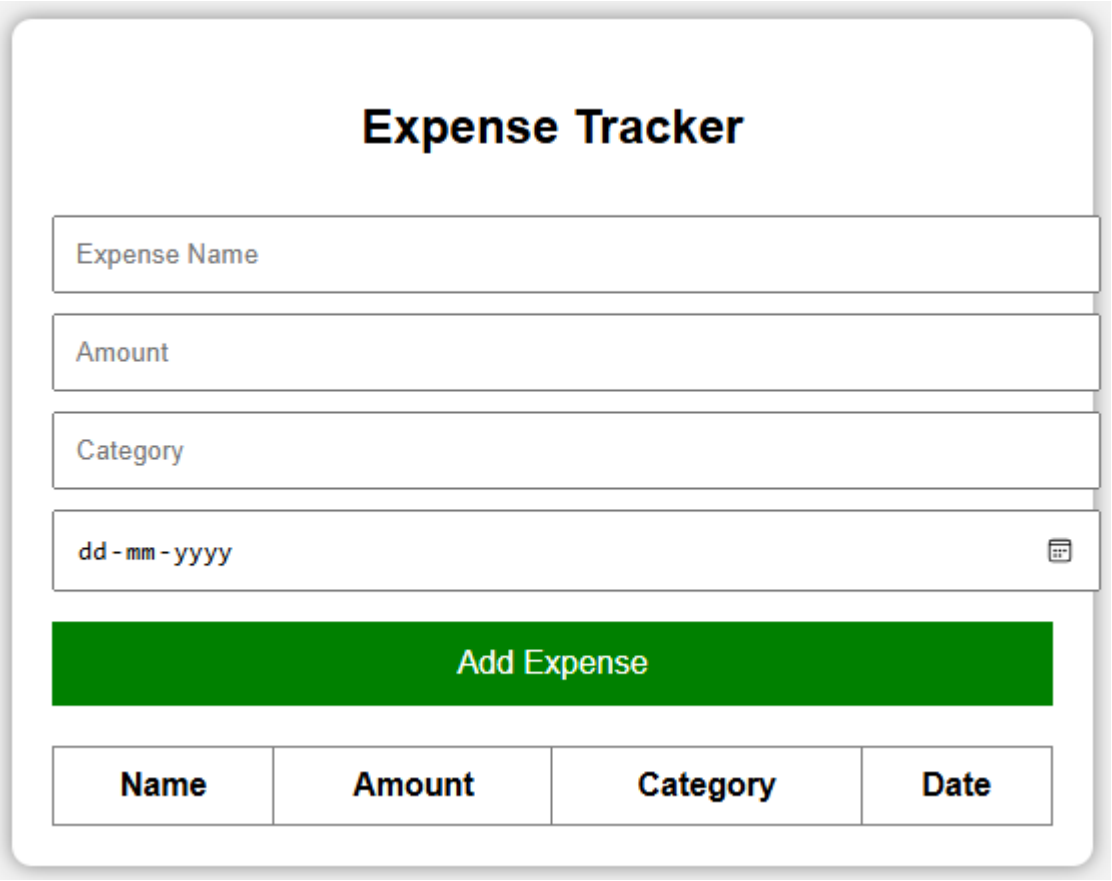


4. S3 Storage Bucket

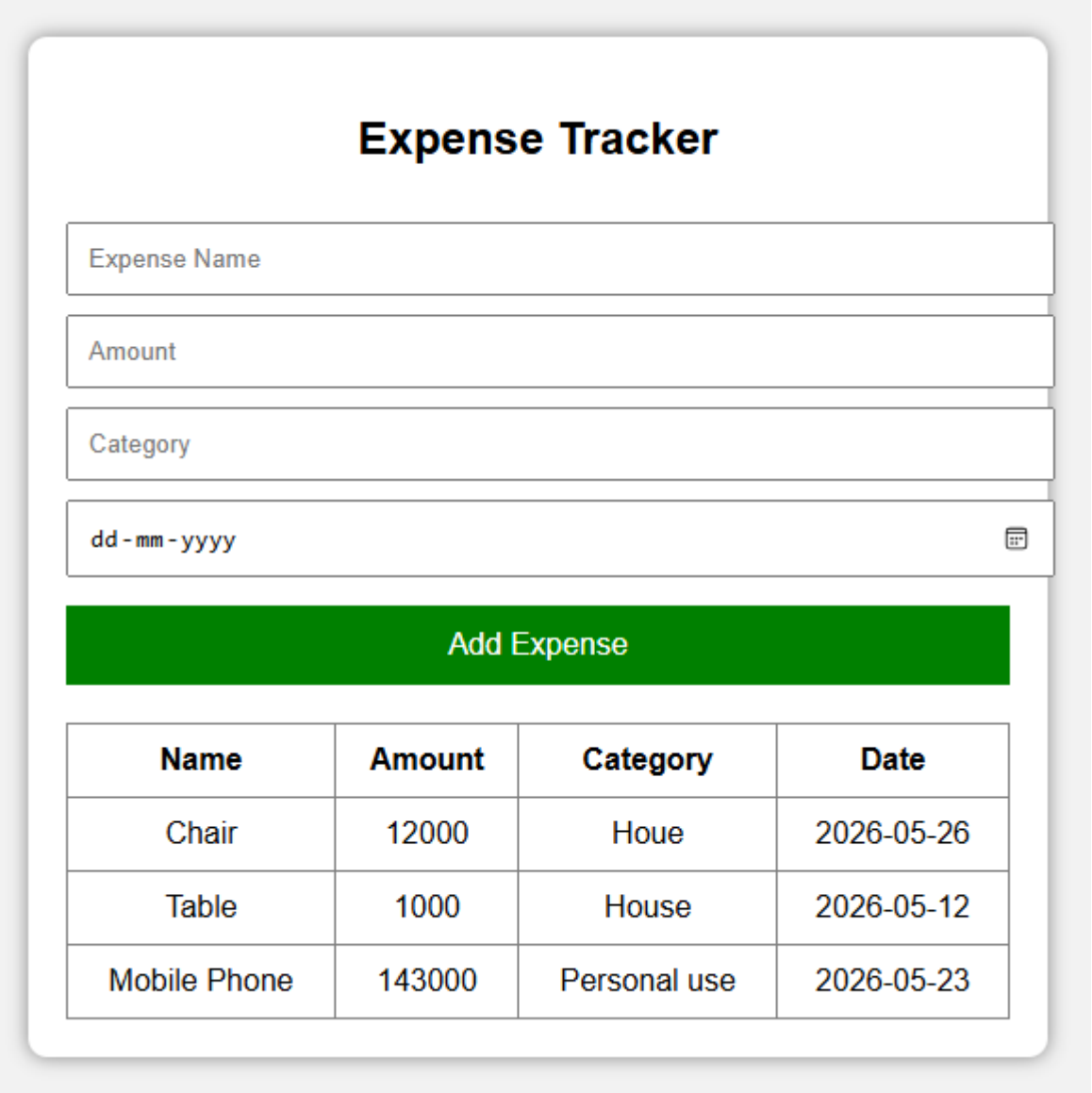
- Create a Static front end web page as public.



5. FRONTEND:



6. OUTPUT



CONCLUSION:

The Serverless Expense Tracker application was successfully developed using AWS cloud services such as **Amazon Web Services, Amazon S3, Amazon API Gateway, AWS Lambda, Amazon DynamoDB,**

The frontend was hosted in S3, API Gateway handled REST API requests, Lambda processed the business logic, and DynamoDB stored the expense details.